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 Studierichting: Informatica
 Oefeningenreeks 2D nr. 10.5

$$\lim_{x \rightarrow -\frac{1}{3}} \frac{6x^2 + 11x + 3}{6x^2 - 7x - 3} = \frac{0}{0}$$

We noemen de vergelijking in de teller en noemer respectievelijk $t(x)$ en $n(x)$

$x_{1,2}$ van $t(x)$:

$$D = 11^2 - 4 \cdot 6 \cdot 3$$

$$= 49$$

$$\sqrt{D} = 7$$

$$x_{1,2} = \frac{-11 \pm 7}{12} \Rightarrow \frac{-1}{3} \text{ en } \frac{-3}{2}$$

$$t(x) = 6(x - x_1)(x - x_2)$$

$$= 6 \left(x + \frac{1}{3} \right) \left(x + \frac{3}{2} \right)$$

$$= (3x + 1)(2x + 3)$$

$x_{1,2}$ van $n(x)$:

$$D = (-7)^2 - 4 \cdot 6 \cdot (-3)$$

$$= 121$$

$$\sqrt{D} = 11$$

$$x_{1,2} = \frac{7 \pm 11}{12} \Rightarrow \frac{3}{2} \text{ en } \frac{-1}{3}$$

$$n(x) = 6(x - x_1)(x - x_2)$$

$$= 6 \left(x - \frac{3}{2} \right) \left(x + \frac{1}{3} \right)$$

$$= (2x - 3)(3x + 1)$$

$$\begin{aligned} & \lim_{x \rightarrow -\frac{1}{3}} \frac{t(x)}{n(x)} \\ &= \lim_{x \rightarrow -\frac{1}{3}} \frac{\cancel{(3x+1)}(2x+3)}{\cancel{(3x+1)}(2x-3)} \end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow -\frac{1}{3}} \frac{(2x+3)}{(2x-3)} \\
&= \frac{\frac{-2}{3} + \frac{9}{3}}{\frac{-2}{3} - \frac{9}{3}} \\
&= \frac{-2+9}{-2-9} \\
&= \frac{7}{11}
\end{aligned}$$

References